

The Alpha–Omega Singularity Coincidence (AOSC): A Global Boundary Framework for Cosmology

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May 23, 2026

Abstract

We introduce the Alpha–Omega Singularity Coincidence (AOSC), a theoretical framework in which the conformal future boundary of a cosmological aeon is identified with the initial singularity of the subsequent aeon. By combining conformal cyclic cosmology with time-symmetric quantum mechanics, the framework replaces arbitrary initial conditions with globally self-consistent boundary constraints. Retrocausal quantum correlations emerge naturally from this structure and act to suppress high-entropy initial states without violating local thermodynamic laws. AOSC offers an alternative resolution of the cosmic entropy problem and provides a structural basis for reinterpreting cosmological initial conditions without invoking inflationary dynamics.

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1 Introduction

The standard Λ CDM cosmological model has achieved remarkable empirical success, accurately describing the large-scale structure of the universe, the cosmic microwave background anisotropies, and the observed accelerated expansion. Despite these successes, the model encounters a foundational difficulty: the origin of the universe’s exceptionally low-entropy initial state.

The so-called Past Hypothesis postulates a highly ordered initial condition but does not provide a physical mechanism for its selection. Inflationary scenarios mitigate some fine-tuning problems but introduce additional fields and parameters, whose ontological status remains uncertain.

The Alpha–Omega Singularity Coincidence (AOSC) proposes a different approach. Rather than treating the initial state as a primitive input, the framework derives it from global boundary conditions imposed across the entire cosmological history.

2 Conceptual and Formal Framework

The Alpha–Omega Singularity Coincidence is formulated as a global, constraint-based cosmological framework in which initial conditions arise from self-consistent boundary relations rather than arbitrary specification.

2.1 Foundational Assumptions

The framework rests on the following assumptions:

1. Physical reality is fundamentally relational.
2. Quantum states may be constrained by both past and future boundary conditions.
3. Only globally self-consistent histories possess non-zero physical amplitude.
4. Classical spacetime and thermodynamic time emerge from deeper quantum structure.

These assumptions preserve local physical laws while extending their global interpretation.

2.2 Two-Boundary Quantum Description

Within a given cosmological aeon, the quantum state of the universe is described using the two-state vector formalism. At an intermediate hypersurface Σ_t , the state is specified by a pre-selected vector $|\psi_{\text{in}}\rangle$ and a post-selected vector $\langle\phi_{\text{out}}|$.

Observable probabilities are given by the Aharonov–Bergmann–Lebowitz rule:

$$P(A) = \frac{|\langle\phi_{\text{out}}|A|\psi_{\text{in}}\rangle|^2}{\sum_i |\langle\phi_{\text{out}}|A_i|\psi_{\text{in}}\rangle|^2}. \quad (1)$$

This formulation introduces correlations that are non-local in time but globally self-consistent.

2.3 Alpha–Omega Singularity Coincidence

A central postulate of AOSC is that the conformal future boundary of a dying aeon (the Omega state) is identified with the initial singularity of the subsequent aeon (the Alpha state).

Under conformal rescaling, all dimensionful quantities lose physical significance near the Omega boundary. This permits a smooth continuation into a new aeon without requiring an anomalously low-entropy initial fluctuation.

The boundary condition may be expressed schematically as:

$$|\psi_{\text{in}}\rangle \equiv |\phi_{\text{out}}\rangle. \quad (2)$$

2.4 Retrocausal Constraints and Entropy

The Alpha–Omega identification induces advanced quantum correlations that propagate backward through the aeon. These correlations act as global constraints on admissible microstates, suppressing high-entropy initial configurations.

Crucially, entropy still increases monotonically within each aeon. The second law of thermodynamics is preserved locally, while the low-entropy origin emerges as a global consistency requirement rather than a brute fact.

2.5 Emergence of Time and Classicality

Time is not treated as a fundamental external parameter. Instead, temporal order emerges from decoherence within globally constrained quantum histories.

Classical spacetime arises when quantum branches satisfy:

1. Stability under environmental decoherence.
2. Compatibility with global boundary conditions.

Histories violating these constraints possess vanishing quantum amplitude.

2.6 Avoidance of Causal Paradoxes

Retrocausal effects do not permit observable causal paradoxes. Only self-consistent histories contribute to the path integral, in accordance with a quantum analogue of the Novikov self-consistency principle:

$$\delta S = \delta \int_{\alpha}^{\omega} \mathcal{L}(\phi, \partial_{\mu}\phi) d^4x = 0. \quad (3)$$

3 Observational Implications

Although AOSC is primarily structural, it suggests potential observational signatures.

3.1 CMB Anomalies

Energy release from black hole evaporation in a previous aeon may appear, after conformal rescaling, as localized low-variance regions in the cosmic microwave background.

3.2 Primordial Gravitational Waves

Gravitational waves may traverse the conformal boundary with minimal distortion. AOSC predicts deviations from smooth, scale-invariant inflationary spectra, potentially observable in future experiments.

3.3 Large-Scale Magnetic Correlations

Retrocausal constraints may generate coherent primordial magnetic fields, correlated with large-scale matter distribution in ways unexplained by standard models.

4 Philosophical Appendix: Time and Boundary

AOSC reframes the arrow of time as a local manifestation of a globally closed boundary condition. The universe does not begin in a special state by accident; it is constrained into coherence by its own future.

Time remains locally causal and directional for observers within an aeon, yet globally forms a self-consistent structure. In this sense, temporal asymmetry is emergent rather than fundamental.

5 Limitations and Open Questions

AOSC is not a complete theory. It lacks a full dynamical realization within quantum gravity and does not yet yield sharply discriminating predictions.

Its value lies in reconfiguring the problem of cosmological initial conditions and suggesting new directions for theoretical and observational inquiry.

6 Conclusion

The Alpha–Omega Singularity Coincidence offers a structural alternative to traditional cosmological narratives. By replacing arbitrary initial conditions with global boundary constraints, it resolves the entropy problem without introducing new fields or modifying local physics.

Future work will focus on explicit toy models, numerical simulations, and connections to candidate theories of quantum gravity.

References

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